# Ryan Chiou

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## Education

**University of Illinois Urbana-Champaign** Expected Dec 2026

*Bachelor of Science in Mechanical Engineering* *GPA: 3.6/4.0*

- **Coursework:** Robotics, Product Design & Additive Manufacturing, Design Optimization, Mechanical Design II

## Experience

**IBM** May 2026 – Aug 2026

*Hardware Engineer Intern | Data & Production Automation* *New York, NY*

- Spearheaded design of a $30K iPhone-based end-of-line inspection rig for IBM Z mainframes to replace a proposed $250K industrial robot, cutting system cost by 88% while maintaining detailed full-view rack coverage
- Designed iPhone mounts and frame fixture in SolidWorks for repeatable imaging, dropping cycle time by 67% to 30 s
- Deployed a Python/PostgreSQL/React dashboard tracking 8-figure material flow to fully automate a 4-hr process

**Tesla** May 2025 – Dec 2025

*Supplier Industrialization Engineer Intern | Battery & Energy Plastics* *Fremont, CA*

- Drove $350K annual savings on injection-molded parts via DFM/DFA, tooling optimization, process development, and material sourcing across Tesla's V4 Supercharger, Powerwall 3, Semi battery, Megapack, and Solar products
- Owned the DFM iteration loop between design release and supplier tooling for 100+ plastic parts, driving rib redesigns, wall thickness tuning, and shrinkage compensation that mitigated warpage, sink, and short-shot risk
- Directed Moldflow analyses of process and boundary conditions to validate melt flow and dimensional conformity
- Root-caused 50+ part defects and drove design/process corrective actions, cutting defect escalations by 95%

**Illini Electric Motorsports** Aug 2023 – Jul 2025

*Engineering Project Lead | Formula SAE Electric* *Urbana-Champaign, IL*

- Led redesign of inverter discharge PCB and energy monitoring sensor enclosure, reducing mass-to-volume ratio by 12% while increasing component capacity, enabling sub-30 s accessibility, and maintaining IP67 ingress protection
- Engineered an accumulator cart with an integrated pawl-and-slot deadman brake mechanism; validated the frame's structural integrity against buckling failure under a 300 lbf load with a FoS of 5 using Ansys FEA
- Optimized undertray venturi profile in Creo using STAR-CCM+ CFD; increased downforce and reduced mass by 9%
- Fabricated 10+ carbon fiber parts via prepreg layups with fixed-volume fractions on CNC-milled polyurethane molds

## Projects

**3D Scanning Robotic Arm** | *Robotics Mechanisms & Controls*

- Designed a 6-DOF arm equipped with a Helios2+ Narrow ToF camera for real-time 3D scanning using point clouds
- Optimized arm segments via a MATLAB scanning workspace model for ∼1 m reach and 1 mm repeatability targets, pairing geared stepper and servo actuators to maximize torque per dollar on a $2K budget
- Implemented forward/inverse kinematics in C++ for teleoperated joint control, driving geared steppers via a Teensy 4.1 and dedicated drivers, and integrated-driver servo actuators over CAN bus
- Developed an Arena SDK/OpenCV/Open3D point cloud pipeline that aligns arm multi-pose scans using forward kinematics and hand-eye calibration, fusing them via ICP into an accurate, unified 3D model

**Single-Input Planetary Walker** | *Robotics Mechanisms*

- Designed a 10-bar linkage leg with path synthesis, tracing a foot path with a 50 mm flat stance and 32 mm lift
- Performed kinematic and dynamic force analysis in Python to size leg link geometry and drivetrain gear ratios
- Packaged robot CAD in Fusion 360 within a 300×180×180 mm envelope and prototyped using FDM 3D printing
- Integrated a Geneva-driven manipulator that sequences payload actuation off a single-DOF motor input

**Deep Sea ROV** | *Underwater Robotics*

- Led the design of a 0.5 kg Delrin ROV frame housing an IP68 electronics enclosure with 6-DOF vectored thrusters
- Integrated Arduino + Raspberry Pi control stack for PWM thruster control and 1080p onboard video over Ethernet
- 3D printed resin camera enclosures, validated by a vacuum hold and 5 m depth submersion at 4× operating duration

## Technical Skills

**Design & Analysis**: SolidWorks, Creo Parametric, Fusion 360, nTopology, Moldflow, Ansys FEA, DFM/DFA, GD&T
**Fabrication**: FDM & Resin 3D Printing, Injection Molding, CNC/Manual Machining, Laser Cutting, Carbon Fiber Layups
**Software & Controls**: Python, MATLAB/Simulink, C/C++, ROS 2, Arduino, Raspberry Pi, SQL, React, JavaScript